

Performance Testing

Date	1 July 2026
Team ID	XXXXXX
Project Name	OptiCrop – Crop Recommendation System
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how your system behaves under expected and peak load conditions. Complete the sections below to document your testing approach, tools used, and results obtained.

Step 1: Testing Overview

Field	Details
Testing Tool Used	Postman, Web Browser (Google Chrome)
Type of Testing	Load Testing
Target Module / API	Crop Prediction Module (/)
Test Environment	Hosted Server (Render Cloud Platform)
Test Date	7 July 2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Access OptiCrop home page	1	30	Home page loads successfully
2	Submit valid soil parameters for crop prediction	5	60	Correct crop recommendation displayed
3	Multiple users perform predictions simultaneously	10	120	System responds without errors
4	Submit invalid or empty input values	5	60	Appropriate validation/error message displayed

Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Avg)	< 2 seconds	1.4 seconds	Pass	Fast response
2	Response Time (Max)	< 5 seconds	3.2 seconds	Pass	Within acceptable limit
3	Throughput (Req/sec)		6 Requests/sec	Pass	Stable performance
4	Error Rate	< 1%	0%	Pass	No errors observed
5	CPU Utilization	< 80%	45%	Pass	Normal usage
6	Memory Utilization	< 80%	52%	Pass	Efficient memory usage

Step 4: Observations & Analysis

The OptiCrop application successfully handled multiple crop prediction requests with stable performance. The average response time remained below the target limit, and no prediction errors were observed during testing. The Flask application and deployed machine learning model responded efficiently, demonstrating that the system is reliable for real-time crop recommendation.

Step 5: Screenshots / Evidence

Attach screenshots of your performance testing tool results below (e.g., JMeter report, Locust graphs, k6 output):

Screenshot Shortcuts x OptiCrop - Crop Recommendation x + Ask Gemini

opticrop-wi6.onrender.com

**OptiCrop**
Smart Agricultural Production Optimization Engine
Machine Learning Accuracy: 99.31%

 Predict Crop

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 Predict Crop

**Recommended Crop for Your Farm**
 **mango**

**Crop Information**
Mango prefers warm climates with moderate rainfall and deep soil.

**Input Summary**

Nitrogen	12.0
Phosphorus	34.0
Potassium	56.0
Temperature	3.0 °C
Humidity	56.0 %
Soil pH	8.0
Rainfall	344.0 mm

 Predict Another Crop

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